Beautiful Music

Symbolic Unconditioned/Conditioned Generation

Overview

- Task 1: Symbolic, unconditioned generation — learn p(x) and sample from it

- Task 2: Symbolic, conditioned generation — harmonization (generate chords following a melody)

- Structure: for each task → EDA & Preprocessing → Modeling → Evaluation → Related Work

TASK 1: SYMBOLIC, UNCONDITIONED GENERATION

Task 1 — Exploratory Analysis & Preprocessing

- **Dataset:** JSB Chorales — 382 Bach chorales, 16th-note resolution
- Each timestep = 4 MIDI pitches (S, A, T, B)
- Pre-split: 229 train / 76 valid / 77 test
- Source: Boulanger-Lewandowski et al. (2012)
- **Preprocessing:** filter 0.4% non-4-note chords, assign voices by register
- **Augmentation:** transpose by [-3, 0, +3, +6] semitones → 916 sequences (4x)

Task 1 — Modeling

- **Formulation:** autoregressive language modeling — $p(x_t | x_{<t})$
- **Objective:** cross-entropy loss (negative log-likelihood) → perplexity
- **Representation:** interleaved SATB tokens [S1, A1, T1, B1, S2, A2, ...]
- **Why LSTM over Transformer?** Small dataset (916 seqs), small vocab (58) — Transformer would overfit
- **Baseline:** Bigram Markov chain — no long-range memory (perplexity 13.59)
- **Model:** 2-layer LSTM, 64-dim embeddings, 128 hidden, ~242K params

Task 1 — Evaluation

- **Perplexity:** Random 58 → Markov 13.59 → LSTM **1.87**

- **Temperature sweep:** T=0.5 (conservative) / 1.0 (balanced) / 1.5 (diverse)

- **Perplexity vs musical quality:** low perplexity ≠ good music — captures local patterns but not phrase structure

Task 1 — Related Work

- Boulanger-Lewandowski et al. (2012) — JSB Chorales benchmark, RNN-RBM
- Eck & Schmidhuber (2002) — first LSTMs for music composition
- Briot, Hadjeres & Pachet (2020) — deep learning for music generation survey
- **Our approach vs prior:** interleaved SATB tokens vs piano-roll encoding

TASK 2: SYMBOLIC, CONDITIONED GENERATION (HARMONIZATION)

Task 2 — Exploratory Analysis & Preprocessing

- **Task:** given soprano melody, generate alto + tenor + bass (harmonization)

- Same dataset, augmentation, and vocab as Task 1

- Soprano is input; other 3 voices are targets

- Soprano-bass heatmap (from EDA) shows structured relationship — contrary motion

- Augmentation helps: same harmonic patterns in different keys, prevents memorization

Task 2 — Modeling

- **Formulation:** sequence labeling — timestep-aligned, not translation

- **Objective:** cross-entropy loss, averaged across 3 voice heads

- **Why BiLSTM?** Harmonization needs future context (cadences, resolution)

- **Why not autoregressive per voice (DeepBach)?** Simpler, faster, but voices can't condition on each other

- **Baseline:** frequency lookup — most common (A, T, B) per soprano pitch (perplexity 125.02)

- **Model:** 2-layer BiLSTM, 3 parallel heads, ~642K params

Task 2 — Evaluation

- **Perplexity:** 125.02 → **5.76**
- **Accuracy:** 41.1% exact / 43.4% pitch-class
- **Voice leading** (parallel 5ths / 8ves, 20 test chorales):
  - Bach: 6 / 5
  - Frequency baseline: 503 / 415
  - BiLSTM: 110 / 115

Task 2 — Related Work

- Ebcioglu (1988) — expert system, 350+ rules, brittle

- Hild et al. (1992) HARMONET — early neural harmonization

- Hadjeres et al. (2017) DeepBach — Gibbs sampling, comparable to Bach in blind tests

- **Our approach vs prior:** predict all 3 voices at once (simpler than iterative), same insight that bidirectional context is key

Discussion + Future Work

- **Worked:** augmentation, interleaved repr, bidirectional context, cosine annealing

- **Limitations:** no phrase structure, parallel motion > Bach, ~41% accuracy

- **Future:** Transformer + music-aware position encodings, voice-by-voice decoding, constrained sampling, chord-label evaluation

Let’s play the music!

1. symbolic_unconditioned.mid (task 1) - generated from scratch at T=1.0
2. symbolic_conditioned.mid (task 2) - real Bach soprano, BiLSTM harmonization